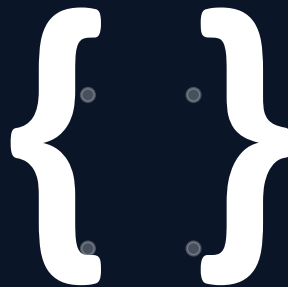


API · Postman · Software Development



REST API

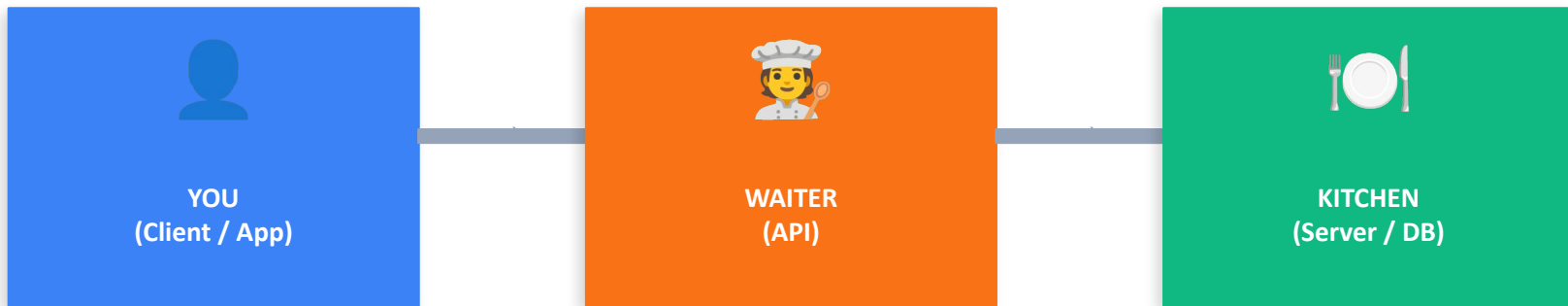
Python requests

Postman

Authentication

What is an API?

API — Application Programming Interface — is a **contract** that defines how software components communicate. It specifies the requests you can make, the data format to use, and the conventions to follow.



Real-world analogy: A waiter (API) takes your order (request) to the kitchen (server) and brings back the food (response) — you never enter the kitchen!

- Abstraction layer between systems
- Language & platform agnostic
- Enables modular software design
- Powers modern web, mobile & IoT

HTTP Methods & Status Codes

HTTP METHODS

GET	Retrieve data GET /users/42
POST	Create resource POST /users
PUT	Update (full) PUT /users/42
PATCH	Update (partial) PATCH /users/42
DELETE	Remove resource DELETE /users/42

STATUS CODES

200	OK – Success	201	Created
400	Bad Request	401	Unauthorized
403	Forbidden	404	Not Found
429	Rate Limited	500	Server Error

Anatomy of a REST API Request

URL BREAKDOWN

`https://api.weather.com/v2/forecast?city=Kathmandu&units=metric`

Base URL

`https://api.weather.com`

Version

`/v2`

Endpoint

`/forecast`

Query Params

`?city=...&units=...`

REQUEST COMPONENTS



Headers

Authorization, Content-Type, Accept, User-Agent



Body (Payload)

JSON data sent with POST/PUT requests



Query String

?key=value pairs appended to the URL



Path Params

/users/{id} — dynamic segments in URL

Python requests Library

```
$ pip install requests
```

```
import requests
# Basic GET request
response = requests.get(
    url=
'https://api.github.com/users/torvalds',
    headers={
'Authorization':
'Bearer YOUR_TOKEN'},
    timeout=
10
)
if response.status_code == 200:
    data = response.
    json()

print(data['name'], data['public_repos'])
```



Simple & Readable

Pythonic API with clean, intuitive syntax



All HTTP Methods

GET, POST, PUT, PATCH, DELETE, HEAD, OPTIONS



SSL Verification

Built-in TLS/SSL support, verify=True by default



Session Support

Persist cookies & headers across requests



Timeouts

Set connection and read timeouts easily



File Uploads

Multipart form data and file streaming

Postman: The API Testing Platform



Request Builder

Build and send HTTP requests visually. Set method, URL, headers, params, and body with a clean GUI.



Collections

Organize related requests into folders. Share collections with your team via Postman workspace.



Environments

Switch between Dev, Staging, Production with variables like `{{base_url}}` and `{{api_key}}`.



Testing

Write JavaScript test assertions. Auto-run tests to validate response structure, status codes, and data.



Automation (Newman)

Run collections from the command line. Integrate into CI/CD pipelines with Newman CLI runner.



Mock Servers

Create mock APIs before the backend is built. Test frontend without a live server.

API Authentication Methods

API Key

Simple token passed as a header or query param. Easy to implement, suitable for server-to-server.

```
Header: X-API-Key: abc123xyz
URL: ?api_key=abc123xyz
```

✓ Simple, fast ✗ No expiry, revocation is manual

OAuth 2.0

Delegated authorization framework. Grants apps limited access without sharing passwords.

```
Flow: Auth Code → Token
Scopes: read:user write:repo
```

✓ Secure, scoped access ✗ Complex to implement

Bearer Token (JWT)

JSON Web Token with encoded claims. Stateless, self-contained. Used in most modern APIs.

```
Authorization: Bearer eyJhbGci...
```

✓ Stateless, expiry built-in ✗ Can't invalidate before expiry

Basic Auth

Base64 encoded username:password. Only use over HTTPS. Legacy, but still common in internal tools.

```
Authorization: Basic
dXNlcjpwYXNz (base64)
```

✓ Very simple ✗ Insecure without TLS

APIs Powering the World

Stripe Payments

Process payments, subscriptions, refunds. Used by millions of businesses globally.

```
POST /v1/payment_intents
```

Google Maps API

Geocoding, directions, places search. Powers ride-sharing, delivery, and navigation apps.

```
GET /maps/api/geocode/json
```

OpenAI API

GPT-4, DALL-E, Whisper. Build AI chatbots, content generators, image tools.

```
POST /v1/chat/completions
```

Twilio SMS/Voice

Send SMS, make calls, WhatsApp. Used for OTP, alerts, customer notifications.

```
POST /2010-04-01/Messages
```

SendGrid Email

Transactional email, marketing campaigns, analytics. 99.99% uptime SLA.

```
POST /v3/mail/send
```

Twitter/X API

Post tweets, search, analyze trends. Used for social listening and automation.

```
GET /2/tweets/search
```

API Best Practices & Error Handling



Rate Limiting

Respect rate limits. Implement exponential backoff + jitter. Use retry-after headers. Cache responses when possible.



Security First

Never hardcode API keys. Use environment variables or secrets managers (Vault, AWS Secrets). Always use HTTPS.



Versioning

Use URL versioning (/v1/, /v2/) or header versioning. Never break existing clients. Deprecate gracefully.



Performance

Request only needed fields (sparse fieldsets). Paginate large results. Use ETags and caching headers.



Error Handling

Always handle 4xx and 5xx. Parse error messages. Implement retries for 429 and 503. Log all failures.



Documentation

Use OpenAPI / Swagger spec. Test docs with Postman. Keep examples real and up-to-date.